

Measurable Signal Intelligence

One decision logic, many industries: the 2i signal-intelligence whitepaper

2i

2026-07-29

Abstract

This whitepaper ties together five technical papers 2i has produced into a single business thesis. Audio fingerprinting, WiFi channel-state-information (CSI) presence and fall sensing, a measured decision map between classical signal processing and deep learning, open-set RF fingerprint verification, and predictive maintenance appear on the surface to belong to five different industries: media, healthcare, telecom, security, and manufacturing. They do not. All five papers run the same underlying mathematics. Physical signals (sound waves, radio-frequency waves, vibration) are reduced to features, and those features are judged against a known, enrolled, or normal state through anomaly detection, entity identification, and open-set classification. The body is one; only the face changes with the industry. That identity is what 2i sells. Not a scattered set of industry point solutions, but one decision logic transplanted across industries.

Contents

1	Abstract	3
2	Why Signal Intelligence: One Decision Logic, Many Industry Faces	3
3	The Gauge Methodology: Measure, Verify, Operate	4
4	Five Studies	4
4.1	Large-Scale Audio Content Fingerprinting	5
4.2	Device-Free WiFi-CSI Presence and Fall Sensing	5
4.3	Classical DSP+GBM vs Deep Learning: A Decision Map for Signal Tasks	5
4.4	Open-Set RF Fingerprint Verification: Separability vs the Operating-Point Gap	6
4.5	Classical-Feature Predictive Maintenance: Vibration and Acoustic	7
4.6	Synthesis Across the Five Tasks	8
5	The Application Map: One Engine, Many Industries	8
6	Honest Boundaries: What Is Demonstrated, What Is at the Edge, What Is Research	9
7	Working With 2i	10
8	References	10

1 Abstract

This whitepaper ties together five technical papers 2i has produced into a single business thesis. Audio fingerprinting, WiFi channel-state-information (CSI) presence and fall sensing, a measured decision map between classical signal processing and deep learning, open-set RF fingerprint verification, and predictive maintenance appear on the surface to belong to five different industries: media, healthcare, telecom, security, and manufacturing. They do not. All five papers run the same underlying mathematics. Physical signals (sound waves, radio-frequency waves, vibration) are reduced to features, and those features are judged against a known, enrolled, or normal state through anomaly detection, entity identification, and open-set classification. The body is one; only the face changes with the industry. That identity is what 2i sells. Not a scattered set of industry point solutions, but one decision logic transplanted across industries.

The five tasks do not sit at the same maturity level, and this whitepaper does not hide that unevenness. Audio content fingerprinting was measured on 999 tracks (GTZAN) and 7,996 clips (FMA-small), reaching 95-99% top-1 identification accuracy under clean conditions and holding 72.5% even under heavy additive noise (SNR -5dB) – mature enough to start a paid pilot today. WiFi-CSI presence/fall sensing and vibration-based predictive maintenance are both verified on real public datasets, but a pilot that re-measures on the client’s own site data is the honest next step. At the far end sits open-set RF fingerprint verification. Separability is real (AUROC 0.87), but at the low false-accept-rate operating point that hard access control actually requires, the true-accept rate collapses to 2-3%, which is not enough evidence to sell it as a single-decision gate. This document keeps these three tiers – what can be sold today, what needs re-measurement, and what is still research – distinct throughout.

That honesty is not a virtue signal; it is a method. 2i works through a gauge methodology that splits every engagement into three gates – Measure, Verify, and Operate – where each gate must earn the budget for the next, and a failing gate is reported honestly as “AI is not the answer here, not yet.” Section 3’s five studies are the evidence that this gauge methodology actually produces defensible numbers, and Section 4’s application map shows how the same decision logic extends into roughly twenty use cases across manufacturing, healthcare, energy, logistics, telecom, and media – without inflating maturity anywhere along the way.

2 Why Signal Intelligence: One Decision Logic, Many Industry Faces

Companies selling AI transformation usually tell an industry-by-industry story: vision inspection for manufacturing, demand forecasting for retail, content recommendation for media. That framing is not wrong, but it misses something. A large share of problems across industries that look different on the surface are mathematically the same question underneath. “Is this sound the track we know?” (audio fingerprinting). “Is someone in this room right now, and did they fall?” (WiFi sensing). “Did this transmission come from the enrolled device it claims to be, or from a device we have never seen?” (RF fingerprinting). “Has this bearing’s vibration drifted outside its normal range?” (predictive maintenance). All four questions reduce to one form: extract features from a physical signal, and decide whether the resulting distribution belongs to a known, enrolled, or normal state.

The thesis 2i argues in this whitepaper is simple. Once this decision logic is built properly once, it does not need to be rebuilt when the industry changes – only the signal source and the feature design need to change. The time-aligned voting structure verified in audio fingerprinting, built to search a long catalog with a short query, becomes the skeleton of RF fingerprinting once the signal type shifts from audio to radio-frequency electromagnetic waves. The approach verified in WiFi-CSI sensing – hand-designed physical features classified with gradient boosting – becomes predictive maintenance once the signal source shifts from wireless channel state to vibration accelerometry. That is the body underneath products that look, from the outside, like unrelated offerings.

There is a dangerous trap in this thesis, and this whitepaper names it directly. “The decision logic is the same” does not mean “the performance is the same.” Section 3’s classical-versus-deep-learning decision map shows that the same decision logic wins with classical methods on tasks with little data and physically interpretable features (modulation classification, interference detection), and loses to deep learning on tasks that are data-rich and

where the task itself is representation learning (co-channel signal separation). Saying the decision logic transfers across industries also means each face of it must be re-verified with real data. This duality – the structure transfers, but the performance must be re-measured task by task – is the method running through this entire document, which is why Section 2 is devoted to how that re-measurement happens in stages, with risk contained at every step.

3 The Gauge Methodology: Measure, Verify, Operate

The most common reason small and mid-sized manufacturers and retailers fail at AI transformation is not a bad model. It is a missing order of operations. When a large contract is signed before anyone has learned what actually works, the money is already spent on things that do not work. 2i inverts that order. Every engagement is split into three gates, each with a pass criterion, and failing a gate means the next gate is not billed.

The first gate is Measure. Over two to three weeks, 2i interviews the site and looks at the data. What is the real problem, does a decision logic on physical signals exist that can solve it, and if so, what does solving it save – answered in numbers. If the conclusion is “AI is not the answer right now,” that is reported honestly. The gate runs at a fixed cost, so the initial commitment stays small.

The second gate is Verify. One candidate from the Measure phase is built and run for real. Not a demo – effect is measured against site data. The core methodological discipline of this whitepaper operates here: a cheap method and an expensive method are set side by side, and the expensive method is only adopted if it actually wins. The result readers will see in Section 3 – classical methods winning three tasks, deep learning winning one, two still undetermined – is exactly this discipline applied across six signal tasks. Only what passes this gate earns the right to be built.

The third gate is Operate. What Verify confirmed is turned into a system that runs in production, and handed to the client’s own team to run it themselves. The pass criterion is not continued dependence on 2i, but a state that runs without 2i.

Underneath all three gates is a measurement discipline. A number produced with a flawed method looks fine until the pilot, where the number collapses along with the client’s trust. 2i applies five safeguards to every experiment log. Pre-registration (pass/fail criteria are written down before the experiment runs, and never moved after seeing the result). Shuffle controls (labels are randomly permuted and the experiment rerun; if performance survives the shuffle, it was statistical noise, not signal). False-alarm-rate-matched comparison (two detectors’ detection rates are only compared once their false-alarm rates are matched – anyone can raise detection rate by lowering the threshold). Leakage prevention (decision thresholds are calibrated only on normal/enrolled data; calibrating on the very targets meant to be caught – unregistered devices, unknown signal sources – uses information that will not exist in the field). And disclosure of negative results (what did not work is reported as not working, not quietly dropped). These five safeguards are not slogans; they are the discipline actually applied in every one of the five papers summarized in Section 3, visible in the confidence intervals and limitation paragraphs attached to every measured number in Section 3 and 4.

2i applies these five safeguards during the Measure gate, and only what survives them spends Verify-gate budget. The single biggest fear in AI adoption – spending money and finding out it did not work – is removed not by better hardware or a better model, but by this procedure. Cutting risk at every gate instead of starting from a gut feeling is how 2i turns AI transformation into a process rather than a gamble. Section 6 returns to what deliverables and quotes these three gates actually produce, and how they combine with government AI-transformation vouchers.

4 Five Studies

This section summarizes five technical papers 2i has measured, each structured as problem, method, headline result, and maturity. Every number is a real measurement; wherever synthetic data was used, that is disclosed at the point it appears. Each subsection links to the full Korean and English PDF of the source paper.

4.1 Large-Scale Audio Content Fingerprinting

Broadcasters, retail platforms, and in-store music services all face one question in common: can the audio coming from this speaker or this broadcast channel be matched to a specific track in a catalog within seconds? Without that ability, royalty settlement, broadcast advertising verification, and in-store music license compliance audits are all impossible. 2i reproduced and measured peak-pair constellation hashing – the algorithm family behind commercial audio-recognition services such as Shazam – on two public datasets.

First, on GTZAN (999 tracks), the goal was simply to verify that the algorithm works at all. Under clean conditions, top-1 identification accuracy reached 95.0% (10-second clips), holding at 72.5% even under strong additive noise (SNR -5dB), and staying at 96.7% across 32-128kbps MP3 compression with almost no degradation. The next question was the real commercialization gate: does accuracy survive as the catalog grows? A cumulative scaling experiment on FMA-small grew the catalog 16x, from 500 to 7,996 songs. Clean top-1 accuracy barely moved, from 99.5% to 99.0%, and stayed at 99.5% under SNR 5dB, while a naive spectrogram-similarity baseline with no time alignment collapsed to 17% under the same condition. Time-aligned voting – not the hash itself – is the real engine of noise robustness.

The most important honest limitation lives in the index. The current index is a single, unsharded, in-memory dictionary; extrapolating per-track memory use linearly implies roughly 868GB of resident memory at a catalog of one million songs. The algorithm does not die at scale, but the current index is pilot-demo grade, and real commercial deployment needs a separate sharded external index layer. This is the most commercially mature of the five tasks – a paid pilot can start today on licensing and API access alone.

Full paper: [Large-Scale Audio Content Fingerprinting](#) (English) - [Korean PDF](#)

4.2 Device-Free WiFi-CSI Presence and Fall Sensing

Falls are among the most common causes of injury in the elderly population, and how quickly a fall is detected in a care facility or a single-occupant elderly household substantially changes recovery outcomes. Cameras raise privacy concerns that make them hard to install in bedrooms and bathrooms, and wearables can be forgotten or knocked off at the moment of the fall. 2i verified, on two public real-world datasets, whether presence, activity, and falls can be distinguished from the channel state information (CSI) of an already-installed WiFi link alone – no camera, no wearable. With no deep learning, 158 hand-designed features (subcarrier variance, Doppler spectrum, envelope statistics) were classified with gradient boosting (LightGBM).

On the UT-HAR benchmark, seven-way activity classification reached 95.2% accuracy, and fall detection reached AUROC 0.990 (91.1% recall at a fixed 0.976 precision). Separately, on real Intel 5300 CSI captures from OPERAnet, motion energy in 10-second windows separated by an order of magnitude between an empty room and walking or sitting/standing activity (52x for walking versus empty, 13x for sitting/standing versus empty). This single statistic alone yielded a presence-detection AUROC of 1.000. Both results are measured on real data.

The paper does not hide its limits. Both datasets come from a single room and a single session, so moving to a new room requires recalibration. It is a repeatedly reported fact in this field that CSI’s statistical character is strongly shaped by furniture layout, wall material, and transceiver placement, and zero-shot transfer of a pretrained model to a new room performs at chance level. Fortunately, gradient boosting needs far fewer labels than deep learning to absorb this recalibration, making site-by-site retraining with a small calibration set practical. In addition, the millimeter-scale chest motion from breathing needed to distinguish a still, present person from an empty room has not yet been verified on public data, and the fall events in UT-HAR are scripted falls by healthy adult volunteers, not the slow, partial falls of the elderly population this use case actually targets. This is a research demonstration of technical feasibility, not a clinically certified medical device.

Full paper: [Device-Free WiFi-CSI Presence and Fall Sensing](#) (English) - [Korean PDF](#)

4.3 Classical DSP+GBM vs Deep Learning: A Decision Map for Signal Tasks

A widely held belief in the wireless signal processing community is that deep learning eventually beats classical signal processing (DSP) and gradient boosting (GBM) on every task. 2i narrowed that belief into a testable

claim and measured it head to head across six tasks – modulation classification, fixed-false-alarm-rate interference detection, drone RF open-set rejection, channel estimation, RF fingerprint fusion, and co-channel signal separation – using the same data, the same splits, and the same metrics for both sides.

The result was not a single verdict; it depended on the task’s structure. On the standard public RadioML2016.10a 6dB subset, classical modulation classification (28-dimensional physical features + LightGBM, accuracy 0.855) clearly beat deep learning (a 1.39M-parameter transformer, 0.681). For interference detection under a fixed false-alarm-rate requirement, classical CFAR (constant-false-alarm-rate) methods held the target false-alarm rate structurally even as the noise floor shifted, while an unconstrained binary deep-learning detector exploded to a false-alarm rate of 1.0 under the same condition. On drone RF open-set rejection, reproduced on the public DroneRF dataset (CC BY 4.0), classical features with Mahalanobis rejection achieved 5.7x the unknown-device rejection rate of deep learning at a single observation ($K=1$: 0.424 versus 0.074).

Deep learning clearly won other tasks. On separating two wireless signals overlapping in the same channel, a learned separator (a Conv-TasNet variant) achieved SI-SDRi +3.01dB, while classical independent component analysis (ICA) scored -14.25dB – worse than the mixed signal itself. Deep-learning embeddings also beat their own classical fusion baseline on RF fingerprint fusion (AUROC 0.9994 versus 0.945 at $K=50$ fusion), though this win comes from a same-session, multi-packet fusion condition that is a research setting, not a commercial product condition. Channel estimation, run through an initial experiment with a known methodological flaw, produced only ambiguous signal, and remains undetermined between classical and deep learning.

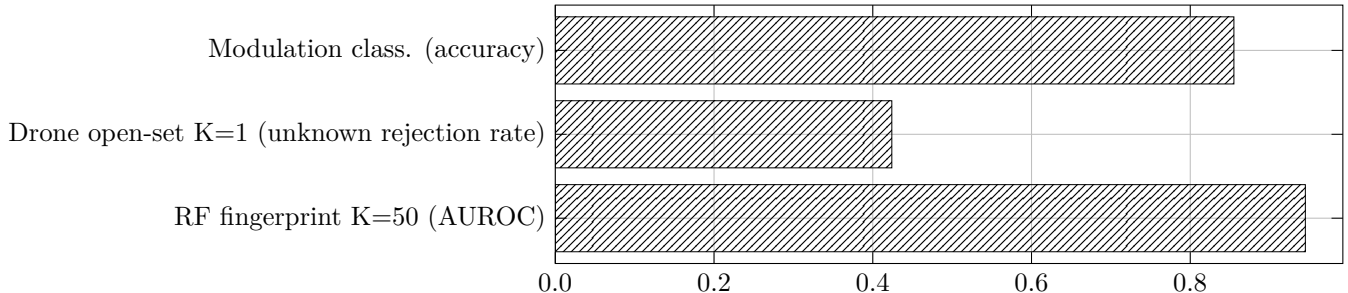


Figure 1: Best measured performance by task: classical vs deep learning

The point of this map is not a scoreboard of “classical wins N, deep learning wins M.” What decided the outcome was task structure. Classical methods won where training data was small and physically interpretable features existed; deep learning won where data was plentiful and the task itself was representation learning. This decision map is the methodological backbone of the thesis in Section 1 – one decision logic wearing different faces across industries means each face’s winning method must be re-verified.

Full paper: [Classical DSP+GBM vs Deep Learning \(English\)](#) - [Korean PDF](#)

4.4 Open-Set RF Fingerprint Verification: Separability vs the Operating-Point Gap

RF fingerprinting identifies wireless devices by the microscopic traces that manufacturing tolerances in the analog transmit chain – oscillator phase noise, ADC/DAC nonlinearity – leave in the radiated waveform. Because it draws an identity signal from the physical layer itself, with no cryptographic key required, it is proposed for use cases such as unregistered-device detection on factory wireless networks and industrial IoT authentication. The problem is that nearly all of these proposals cite accuracy on a closed-set question – which of the known devices sent this – when access control actually faces a different, open-set question: can an unregistered device, attempting to impersonate an enrolled identity, be rejected at a low false-accept rate?

2i measured this question across three public datasets (WiSig, INRIA PLA, WIDEFT). Single-packet ($K=1$) AUROC ranged 0.70-0.87 across the three, clearly above chance (0.5). Separability is real. But moving to hard authentication – the low false-accept-rate operating point ($TAR@FAR$) an access-control gate must actually

decide at – causes this signal to collapse. On INRIA (CC BY 4.0, commercially usable), true-accept rate (TAR) was 20.8% at FAR=1%, and only 2.9% at FAR=0.1%.

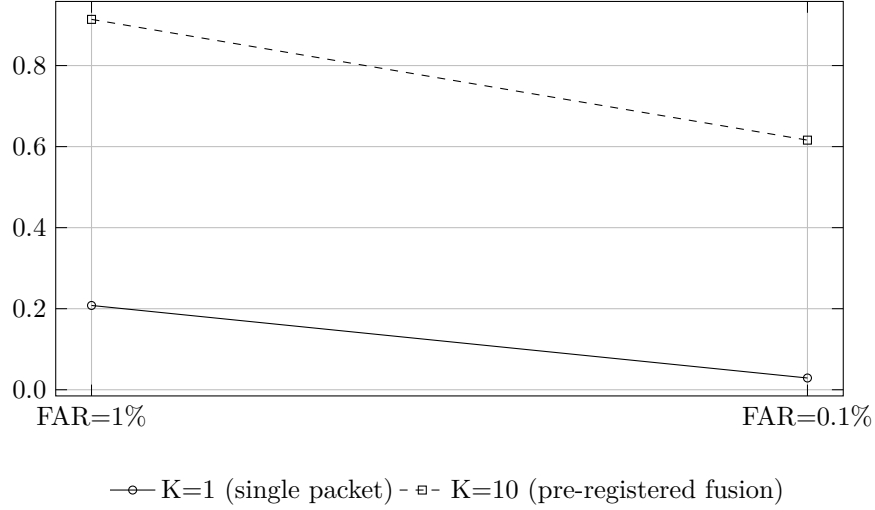


Figure 2: RF fingerprinting: TAR collapses as FAR tightens

Fusing ten pre-registered packets ($K=10$) substantially narrows this gap. At FAR=1%, TAR climbs to 91.4%, close to an access-control threshold, but tightening FAR to 0.1% drops TAR back down to 61.6%. Even after accumulating ten packets, the gap at the strictest operating point does not fully close. The fact that this same gap reproduces across two datasets with different licenses (WiSig is non-commercial only, INRIA is commercially usable) suggests this is a property of the task itself, not an artifact of one dataset.

The conclusion is that with today’s public data and public methods, there is not enough evidence to stand up RF fingerprinting as a single-decision hard access gate. What 2i can honestly sell is not a single-decision gate but a continuous-valued risk-score layer that gets fused with other signals. The consumer of that confidence score should be a fusion platform, not the end user directly, and a low-confidence attempt should route to additional verification (secondary authentication, an administrator alert) rather than an immediate block – a defense-in-depth design matched to the current evidence level.

Full paper: [Open-Set RF Fingerprint Verification](#) (English) - [Korean PDF](#)

4.5 Classical-Feature Predictive Maintenance: Vibration and Acoustic

Predictive maintenance – catching rotating-equipment faults (bearings, pumps, motors) early – was measured with classical signal-processing features and gradient boosting rather than deep learning. Envelope spectrum, cepstrum, and wavelet band-energy features – all physically grounded – were extracted and classified with LightGBM, and compared against a baseline using only simple time-domain statistics.

Vibration signals were real 12kHz accelerometer measurements from the public standard bearing-fault dataset CWRU (Case Western Reserve University Bearing Data Center). On clean data, the full feature set and the simple baseline barely differed (AUROC 1.000 versus 0.998) – a large fault is easy for anyone to catch. The real test was noise. Lowering signal-to-noise ratio to -5dB, the full feature set held at 0.636 while the simple baseline collapsed to 0.186 – physically grounded features degrade far more gracefully under noise. Separately, the same fixed-false-alarm-rate CA-CFAR detector used in industrial wireless interference detection was verified on this domain, holding a target false-alarm rate of 5% to a measured 5.29% while achieving a 98.0% detection rate.

Acoustic signals are the one place in this whitepaper where synthetic, not measured, data was used, because real factory microphone recordings have not yet been obtained. Fault orders were computed from a standard nine-ball deep-groove bearing geometry and generated procedurally; no CWRU data or recordings were used. The same pattern reproduced on this synthetic data: accuracy fell from 90.3% clean to 49.2% at -5dB for the full feature set, while the simple baseline started low and barely moved (31.2% to 30.2%). But the fact that the pattern reproduced does not make the acoustic numbers real measurements.

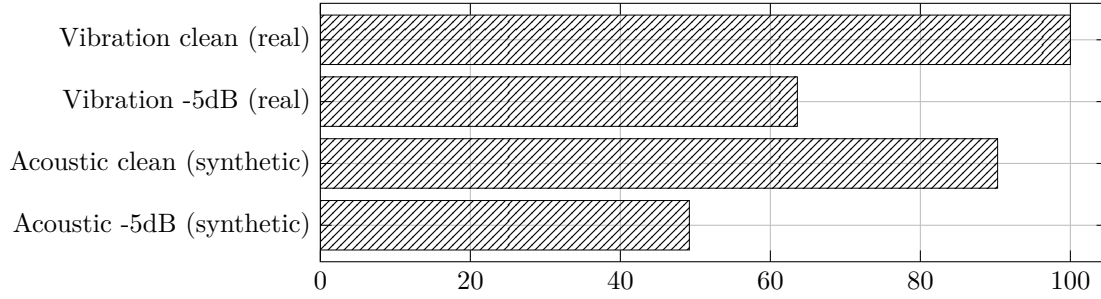


Figure 3: Performance under low SNR

2i does not hide this distinction. Vibration is measured and is a deployable pilot candidate. Acoustic is synthetic, and verification on the real industrial acoustic public dataset MIMII is the next step. Sharing a methodology is a different claim from having completed verification, and this paper does not conflate the two.

Full paper: [Classical-Feature Predictive Maintenance \(English\)](#) - [Korean PDF](#)

4.6 Synthesis Across the Five Tasks

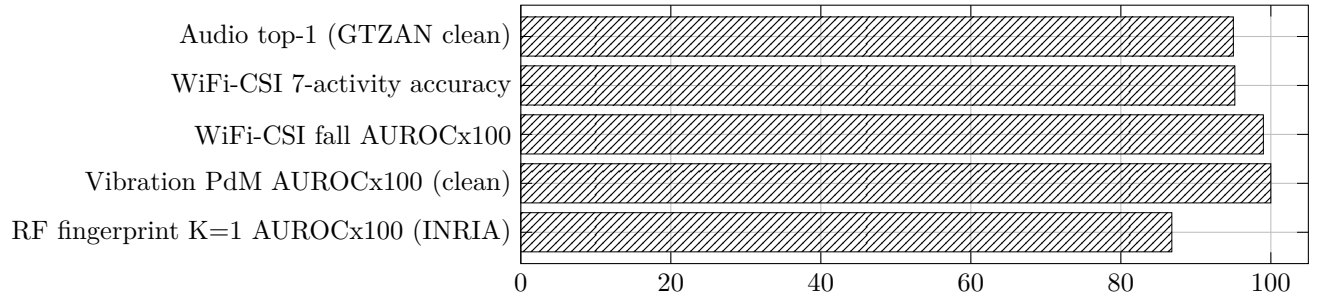


Figure 4: Headline measured metric across five tasks

What this figure does not say matters. Five bars of similar height do not mean five tasks of equal commercial maturity. Audio fingerprinting holds 72.5% even under harsh conditions and can be sold today; RF fingerprinting shows the same 86.8% separability collapsing to 2.9% at the hard authentication operating point. The real measure of maturity is not the height of the clean bar but how well that height holds under harsh conditions and hard operating points – the boundary Section 5 returns to.

5 The Application Map: One Engine, Many Industries

The decision logic in Section 3 – extracting features from a physical signal and judging deviation from a known, enrolled, or normal state – does not stop at telecom and media. Below is an application map drawn from technology 2i has actually measured or holds. The tags are not inflated. **Demonstrated** means already built and measured. **Extension** means it assembles naturally from technology already held, but needs re-verification on the client’s own data. **Research** means the method exists but verification is not yet complete. **Concept** means it is an idea stage where experiments have not yet started.

The top six priorities first: audio content ID (media/broadcast, **Demonstrated**), rotating-equipment predictive maintenance (manufacturing, **Demonstrated**), presence detection for energy optimization (smart buildings, **Extension**), unregistered-device detection on OT networks (manufacturing, **Extension**), drone RF detection/identification (security/defense, **Research**), and early partial-discharge warning in substations (energy, **Concept**). The first two can start a paid pilot today; the middle two start from a client-data pilot; the last two are honestly still research and idea stage.

Manufacturing. Detecting a factory wireless interference source has already been demonstrated (catching the fact that an interferer exists); pinpointing its exact location is the next Extension task. Rotating-equipment predictive maintenance (acoustic/vibration) transplants Section 3.5’s demonstration directly, catching early bearing/pump anomalies through noise and vibration. Detecting unregistered devices on OT networks – finding devices quietly plugged onto a production line via RF fingerprint – is an Extension.

Healthcare. Detecting interference and anomalies in hospital wireless networks (patient monitors, IoMT) without interrupting the ward is an Extension. Contactless presence/fall monitoring (patient rooms, elder care) directly transplants Section 3.2’s UT-HAR/OPERAnet demonstration and is an Extension, with on-site hospital validation as the next step. Contactless vital-sign (respiration/heartbeat) radar sensing is Research, and requires clinical validation and medical-device certification as separate gates.

Smart Buildings and Homes. Contactless presence detection for HVAC and lighting energy optimization is already a commercial technology family deployed at tens of millions of households abroad, and 2i reproduced the same principle on public data (UT-HAR) at AUROC 0.99 (Extension). Fall detection for elderly single-occupant households rests on the same Extension-level evidence. Sweeping for unauthorized wireless devices hidden in meeting rooms is Research.

Energy and Utilities. Early partial-discharge warning for substations and transformers is a detection principle that has been an industry standard for decades, but transplanting 2i’s fixed-false-alarm-rate detector onto this signal has not yet started, hence Concept. Detecting anomalies and tampering on smart-grid metering networks is Research.

Logistics and Cold Chain. RF/BLE tag authenticity verification (GPS tells you a location, not whether the tag itself is genuine) and detecting unregistered beacons in warehouses and yards are both Concept-stage, ideas to be verified together.

Security and Defense. Unknown signal-source identification and spectrum situational awareness is an Extension that broadens the open-set identification engine into a general spectrum-surveillance capability. Unauthorized drone detection/identification rests on the $K=1$ measurement from Section 3.3 (unknown rejection rate 0.424), but the absolute number is still low, so it is marked Research – while noting that a commercial anti-drone market already exists. Sweeping facilities for illicit transmitters is also Research.

Telecom and Spectrum. An interference-detection SDK for SDR/RF vendors (including O-RAN and private-5G slice isolation) has been measured on synthetic data at a 98.0% detection rate and 5.3% false-alarm rate, and is Demonstrated. CBRS-style spectrum co-existence monitoring is an Extension.

Media and Broadcast. Audio content ID and broadcast copyright monitoring is Demonstrated, backed directly by Section 3.1’s measurements, and is the item in this map where a paid pilot can start soonest.

Agriculture. Early acoustic detection of respiratory illness in swine barns is a cross-domain bet at some distance from 2i’s core RF/signal stack – only the method transfers, and it is Concept-stage.

Retail. In-store WiFi-CSI foot-traffic and congestion analysis (camera-free) applies Section 3.2’s presence-decision logic and is Concept-stage.

What this map shows is not roughly twenty different products but one extension surface: the decision logic verified across five papers, transplanted across signal sources (audio, WiFi, RF, vibration, acoustic) and domain-specific feature design. The distribution of tags – roughly four Demonstrated, eight Extension, six Research, five Concept – is itself a statement of where 2i stands today. The core engine is real and verified; most industry transplants still sit in Extension, Research, or Concept.

6 Honest Boundaries: What Is Demonstrated, What Is at the Edge, What Is Research

The whole of this whitepaper compresses into one sentence: the decision logic is singular, but the weight of evidence differs by task, and 2i does not hide that weight.

Demonstrated. Audio content fingerprinting held 95-99% under clean conditions and 72.5% under harsh conditions (SNR -5dB) across measured data from 999 and 7,996 tracks. WiFi-CSI presence/fall sensing measured

95.2% seven-activity accuracy on 4,973 UT-HAR windows and AUROC 1.000 for presence detection on real Intel 5300 CSI captures from OPERAnet. Vibration-based predictive maintenance held AUROC 1.000 clean and 0.636 at -5dB (3.4x the baseline) on CWRU’s real accelerometer data. These three axes are the measured foundation on which a paid pilot, or a client-data re-verification PoC, can start today.

At the edge. Open-set RF fingerprint verification confirmed real separability (AUROC 0.87) but collapses at the low-false-accept-rate operating point hard authentication requires (TAR at FAR=0.1% is 2.9%). This is not a failure to be hidden; it is an edge that forces an honest redesign of the product form. 2i crosses this edge with a different product shape – a risk-score layer instead of a single-decision gate – and several of Section 4’s Extension and Research items, such as OT-network unregistered-device detection and internal illicit-transmitter sweeps, stand on this same edge.

Research and concept stage. The acoustic axis of predictive maintenance has only been verified on procedurally synthesized data, with reproduction on a real industrial acoustic dataset (MIMII) as the next step. Channel estimation remains undetermined between classical and deep learning. Partial-discharge early warning, acoustic swine-illness detection, and contactless vital-sign radar sensing are cases where the detection principle itself is an industry standard or verified in another field, but transplanting 2i’s decision logic onto that specific signal has not yet started, or has only just begun.

These three tiers sit side by side so 2i does not claim to sell a finished, certified product across the board. A company that says plainly “only audio content ID is sellable today” is, over the long run, a more trustworthy partner than one that inflates the other four into “almost there.” That is exactly why a free diagnostic consultation exists – to establish, before any contract, whether a client’s problem sits in the Demonstrated tier, at the edge, or still needs research.

7 Working With 2i

The three gates introduced in Section 2 are not an abstract methodology; each one leaves a concrete deliverable. Finishing the Measure gate hands the client a problem statement, a shortlist of candidate applications (in the format of Section 4’s map), and a diagnostic report with expected effect and cost for each candidate. Finishing the Verify gate hands over a performance table measured on the client’s own site data (in Section 3’s format – a cheap method and an expensive method side by side) and an explicit recommendation on whether to scale. Finishing the Operate gate hands over an operating manual, alongside a system the client’s own team can run without 2i.

This sequence fits structurally with South Korea’s 2026 government AI-transformation support program, which funds up to KRW 50 million per demand company: KRW 10 million for expert consulting (diagnosis and design), and KRW 40 million for PoC verification (the Verify stage). The Measure gate’s output becomes the basis of the grant application itself, and the Verify gate becomes the PoC track record the voucher requires. As the supplying (hub) company, 2i handles the application paperwork end to end, and designs the technical plan and the funding application together so that a substantial share of transformation cost can be covered by the support program. Support amounts and eligibility follow each year’s official announcement.

Measure, verify, operate. Cutting risk at gates instead of starting from a guess, and attaching a real measured number and an honest limitation to every gate along the way – that is what this whitepaper’s five studies were meant to demonstrate about how 2i sells AI transformation. A free diagnostic consultation is where it starts.

8 References

1. 2i. Large-Scale Audio Content Fingerprinting: Robust Song Identification Under Noise, Compression, and Tempo Shift, and the Honest Limits of Catalog Scaling. 2026. GTZAN (999 tracks), FMA-small (8,000 clips, Creative Commons).
2. 2i. Device-Free WiFi-CSI Presence and Fall Sensing: A Real-Data Verification. 2026. UT-HAR/SenseFi benchmark (Yousefi et al. 2017; Yang et al. 2023), OPERAnet (Bocus et al. 2022, CC0).
3. 2i. Classical DSP + GBM vs Deep Learning: A Measured Decision Map for Signal Tasks. 2026. RadioML2016.10a, DroneRF (CC BY 4.0).

4. 2i. Open-Set RF Fingerprint Verification: Separability vs the Operating-Point Gap. 2026. WiSig (CC BY-NC-SA 4.0), INRIA PLA (Zenodo 18268648, CC BY 4.0), WIDEFT (Zenodo 4116383, CC BY 4.0).
5. 2i. Classical-Feature Predictive Maintenance: Vibration and Acoustic. 2026. CWRU Bearing Data Center (real vibration), procedurally synthesized acoustic data (MIMII verification planned).
6. Wang, A. (2003). An Industrial Strength Audio Search Algorithm. ISMIR 2003.
7. Yousefi, S., Narui, H., Dayal, S., Ermon, S., & Valaee, S. (2017). A Survey on Behavior Recognition Using WiFi Channel State Information. IEEE Communications Magazine.
8. Bocus, M. J., Li, W., Vishwakarma, S., et al. (2022). OPERAnet: A Multimodal Activity Recognition Dataset Acquired from Radio Frequency and Vision-Based Sensors. Scientific Data.